

Exact Kolmogorov Ratios under Self-Similar Focusing

Michael Grafiel S Puno

August 2026 (v0.1.0)

Michael Grafiel S Puno

August 2026 (v0.1.0)

Abstract

We compute exactly the behavior of the Kolmogorov functional

$$K[u] = \|u\|_{\infty} / \epsilon^{1/3}, \quad \epsilon = \nu \int |\operatorname{grad} u|^2 dx,$$

under the concentration scenarios relevant to possible blowup of the 3D Navier-Stokes equations. For Gaussian spikes we obtain the exact law $K = (2Aw/(\nu \sqrt{\pi}))^{1/3}$, where A is amplitude and w is width: the ratio depends on spike mass, not sharpness, and sharpening at fixed amplitude DRIVES THE RATIO TO ZERO. Under self-similar focusing $u = s^{-\sigma} F(x s^{-\sigma})$, $s = T - t$, in d dimensions, the ratio evolves as a pure power law $K(s) = K(1) s^{-\sigma(d-1)/3}$. At the scale-invariant rate $\sigma = 1/2$ this recovers the exponent $-(d-1)/6$; in particular $d = 1$ makes the ratio an invariant of ALL power-law focusing rates simultaneously, which explains structurally why one-dimensional global regularity is provable. In $d = 3$ every self-similar candidate -- including all type-II rates excluded from existing nonexistence results only at $\sigma = 1/2$ -- violates the Kolmogorov bound at a computable polynomial rate that GROWS with σ : the exclusion landscape is monotone. All exponents are verified numerically to machine precision ($\sim 1e-15$) across 12 cases; all static laws across 285 cases. We state plainly what this does and does not prove: it quantifies, uniformly over the entire self-similar family, the gap that the open inequality $\|u\|_{\infty} \leq C \epsilon^{1/3}$ must close for evolving solutions.

1. Introduction

The millenium regularity problem for 3D Navier-Stokes is equivalent, via the reduction in [1], to the uniform-in-time inequality

$$\|u(t)\|_{\infty} \leq C \epsilon(t)^{1/3}, \quad (1)$$

where $\epsilon(t) = \nu \int |\operatorname{grad} u|^2 dx$ is the instantaneous dissipation. This note does not prove (1). It computes, exactly and with explicit constants, how the ratio $K[u] = \|u\|_{\infty} / \epsilon^{1/3}$ behaves under the concentration profiles and self-similar evolutions that any blowup scenario must resemble. Three results emerge:

R1 (exact spike law). Static Gaussian spikes obey $K = (2/\sqrt{\pi})^{1/3} (Aw/\nu)^{1/3}$: dissipation punishes concentration. Sharper spikes are MORE subcritical.

R2 (divergence rates). Self-similar focusing at rate σ drives K to infinity like $s^{-\sigma(d-1)/3}$. The target inequality closes a polynomial gap -- slow, explicit, monotone in the focusing rate.

R3 (dimension-one invariance). In $d = 1$ the exponent vanishes for EVERY $\sigma \geq 1/2$. No power-law scenario diverges K . This is the structural reason Theorem A of [1] (1D global

regularity) is provable while its 3D analogue is the millennium problem.

Everything here is elementary dimensional analysis elevated by exact constants and machine-precision verification; we claim novelty only in the packaging, the constants, and the uniform two-parameter (σ , d) treatment, which we have not found stated elsewhere [6].

Conventions: $E = (1/2) \int |u|^2 dx$, $Z = (1/2) \int |\text{grad } u|^2 dx$, $\text{eps} = -dE/dt = 2 \nu Z$. All norms are computed on grids with 161 points per axis; derivatives by central differences.

2. The Exact Spike Law (Static)

Let $u(x) = A \exp(-x^2 / (2 w^2))$ in one dimension. Then

$$\begin{aligned} \|u\|_{\infty} &= A \\ \int (u')^2 dx &= A^2 \sqrt{\pi} / (2w) \\ \text{eps} &= \nu A^2 \sqrt{\pi} / (2w) \end{aligned}$$

and therefore

$$\begin{aligned} K &= A / \text{eps}^{1/3} \\ &= (2 A w / (\nu \sqrt{\pi}))^{1/3} \\ &= (2/\sqrt{\pi})^{1/3} * (Aw / \nu)^{1/3}. \end{aligned} \quad (2)$$

Numerical verification across 48 parameter sets spanning amplitudes A in $\{1, \dots, 50\}$, widths w in $\{0.05, \dots, 0.5\}$ (sharpness up to 1000), and viscosities ν in $\{0.01, 0.05, 0.1\}$: the measured ratio $K / (Aw/\nu)^{1/3}$ equals

$$1.041 \pm 0.000 \quad (\min = \max)$$

against the analytic constant $(2/\sqrt{\pi})^{1/3} = 1.04108$. Spot check: $A = 50$, $w = 0.5$, $\nu = 0.01$ predicts $K = 14.130$; simulation gives 14.130.

The physical content of (2): K depends on the MASS $m = A w$, not the sharpness $s = A/w$. At fixed amplitude, taking $w \rightarrow 0$ sends $K \rightarrow 0$. Mechanism: concentrating into width w costs enstrophy $\sim A^2/w$, hence $\text{eps} \sim \nu A^2/w$, which outruns the numerator in the ratio. Viscosity beats focusing. This is the quantitative form of the outward-cascade picture: energy cannot pile up because the dissipation cost of concentration diverges faster than the concentration itself.

Extended verification (285 cases): radial 3D flows (180), axial 3D dipoles (9), spectral 1D random-phase fields (48, with the L2 interpolation Gap 1 of [1] holding 48/48), and Gaussian spikes (48). All bounded, all consistent with (2).

3. Type-I Focusing Rate

Type-I self-similar blowup has the scale-equivariant form

$$u(x, t) = s^{-1/2} F(x s^{-1/2}), \quad s = T - t,$$

corresponding to $\lambda(t) = s^{-1/2}$ in the NS symmetry $u \rightarrow \lambda u(\lambda x, \lambda^2 t)$. With $y = x s^{-1/2}$, $dx = s^{d/2} dy$:

$$\begin{aligned} \text{grad } u &= s^{-1} \text{grad } F, \\ Z &= s^{d/2 - 2} Z_F, \\ \|u\|_{\infty} &= s^{-1/2} \|F\|_{\infty}, \end{aligned}$$

so

$$K(s) = K(1) * s^{(1-d)/6}. \quad (3)$$

Verification (separable Gaussians, 13 log-spaced times s in $[1e-4, 1]$, per dimension):

d=1:	measured slope	-0.0000	predicted	0.0000	error	1.8e-16
d=2:	measured slope	-0.1667	predicted	-0.1667	error	2.8e-16
d=3:	measured slope	-0.3333	predicted	-0.3333	error	3.3e-16

Cross-check in $d = 1$ against the exact law (2): self-similar value 4.8354 vs analytic 4.8323 -- 0.07 percent difference, attributable entirely to grid discretization.

Two readings of (3):

- $d = 1$: K is an INVARIANT of type-I focusing. Nothing diverges. Consistent with proved 1D global regularity [1].
- $d = 3$: K diverges like $(T-t)^{-1/3}$. Slowly. Polynomially. The inequality (1) would forbid exactly this, and independent consistency holds: Necas-Ruzicka-Sverak [3] and Tsai [4] exclude nonzero type-I self-similar blowup outright.

4. Type-II Generalization and Logarithmic Corrections

Since $\sigma = 1/2$ is excluded, any self-similar blowup must be type-II: faster than scale-invariant. Consider the generalized family (exactly NS-scale-equivariant with $\lambda = s^{-\sigma}$)

$$u(x, t) = s^{-\sigma} F(x s^{-\sigma}), \quad \sigma \geq 1/2.$$

With $y = x s^{-\sigma}$, $dx = s^d dy$:

$$\begin{aligned} \text{grad } u &= s^{-2\sigma} \text{grad } F, \\ Z &= s^{\sigma(d-4)} Z_F, \\ ||u||_{\infty} &= s^{-\sigma} ||F||_{\infty}, \end{aligned}$$

hence

$$K(s) = K(1) * s^{-\sigma(d-1)/3}. \quad (4)$$

Verification across σ in $\{0.5, 0.75, 1.0, 1.5\}$ and d in $\{1, 2, 3\}$ (12 cases):

d=1:	slope	0.0000	for EVERY σ ,	errors	$\leq 2e-16$,
	growth factor	1.00	everywhere.		
d=2:	slopes	-1/6, -1/4, -1/3, -1/2;	errors	$\leq 4e-16$.	
d=3:	slopes	-1/3, -1/2, -2/3, -1;	errors	$\leq 8e-16$.	

Logarithmic corrections. If the focusing factor carries a slowly varying multiplier $L(s)$, $\lambda(s) = s^{-\sigma} L(s)$, then to leading order in slowly-varying approximation the ratio gains a factor $L^{(d+1)/3}$. In $d = 3$ this permits e.g.

$(\log(1/s))^{\gamma(d+1)/3}$ -type growth; in $d = 1$ it permits only $(\log(1/s))^{2/3}$ -- logarithmic, never polynomial. The $d = 1$ invariance is robust to log corrections in precisely the sense that matters: no self-similar scenario can make K diverge as a power law in one dimension.

5. The Monotonicity of Exclusion

Combine (4) with the existing nonexistence results:

- Type-I ($\sigma = 1/2$): EXCLUDED unconditionally in the settings of [3, 4].
- Type-II ($\sigma > 1/2$): divergence rate $-\sigma(d-1)/3$ exceeds the type-I rate $-(d-1)/6$ in absolute value for every $d > 1$. Any surviving candidate focuses harder and therefore violates (1) MORE strongly.
- Discrete self-similarity (λ -DSS): existence of nonzero

backward DSS blowup remains open [5]; our continuity-based rates do not settle it, but the static spike law (2) still applies pointwise along any concentrating sequence.

In words: the harder a solution tries to focus, the larger and more polynomially explicit the violation of the Kolmogorov bound becomes. One inequality, (1), excludes the ENTIRE continuous self-similar family simultaneously, with margin growing at the computable rate $\sigma(d-1)/3$.

6. What This Does Not Prove

Stated plainly:

1. Results (2)-(4) concern PROFILES and ANSATZE, not solutions. They do not establish that evolving Navier-Stokes solutions satisfy (1).
2. The exponent $-(d-1)/3$ is dimensional analysis. Its value here lies in the exact constants, the machine-precision checks, the uniform (σ, d) treatment, and the structural explanation of why $d = 1$ sits exactly at the boundary of provability.
3. Discretely self-similar blowup is untouched beyond the static law.

What the computation buys: the millennium gap is now quantified as a single slow power law, uniform over the whole self-similar family, with the dimension-one case exactly critical. Any future attack on (1) knows precisely which polynomial margin it must recover.

References

- [1] M. Grafiel S Puno, The Navier-Stokes Millennium Problem: Reduction to Kolmogorov Theory via the Tautology Principle, preprint (2026); sections 4, 5.1-5.3.
- [2] J. Leray, Sur le mouvement d'un liquide visqueux remplissant l'espace, Acta Math. 63 (1934), 193-248.
- [3] J. Necas, M. Ruzicka, V. Sverak, On Leray's self-similar solutions of the Navier-Stokes equations, Acta Math. 176 (1996), 283-294.
- [4] T.-P. Tsai, On Leray's self-similar solutions of the Navier-Stokes equations satisfying local energy estimates, Arch. Rational Mech. Anal. 143 (1998), 29-51.
- [5] Z. Bradshaw, P. Phelps, Asymptotic properties of discretely self-similar Navier-Stokes solutions with rough data, arXiv:2409.13586 (2024).
- [6] L. Caffarelli, R. Kohn, L. Nirenberg, Partial regularity of suitable weak solutions of the Navier-Stokes equations, Comm. Pure Appl. Math. 35 (1982), 771-831.
- [7] A. N. Kolmogorov, The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers, Dokl. Akad. Nauk SSSR 30 (1941), 301-305.

Companion verification scripts: experiments/outward_cascade.py, outward_cascade_extended.py, selfsimilar_cascade.py, selfsimilar_type2.py; data outputs in data/*.json.